

International Cybersecurity and Digital Forensics Academy (ICDFA)

HUSSEINI SULEIMAN

2025/FWSD/11478

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DATE: 15TH September, 2026

SBT-DF203-Lab5_2025/FWSD/11478_Husseini Suleiman

Address	HWtype	HWaddress	Flags Mask	Iface
192.168.142.254	ether	00:50:56:fc:94:0d	C	eth0
192.168.142.2	ether	00:50:56:f4:d5:0d	C	eth0

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 5: ARP Poisoning Forensics

Three-Week Delivery Block 2/3 | 12-18 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 5
Lab Title	ARP Poisoning Forensics
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	arp.pcap and an isolated host-only ARP capture

1. Source Materials and Slide Alignment

- Address Resolution Protocol Forensics: ARP operation, ARP caches, request/reply fields, capture with tshark, ARP spoofing/poisoning, man-in-the-middle consequences and evidence analysis.
- The slide practical compares clean and poisoned ARP tables and uses a supplied arp.py script only within the same LAN.
- The assignment requires legitimate ARP capture, spoofing capture and proof of a poisoned ARP table.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A user reports that the default gateway intermittently resolves to the wrong MAC address. You will first document normal ARP resolution, then analyze a known ARP-poisoning capture. In an instructor-approved host-only network, you may reproduce a short controlled poisoning event, capture the evidence, restore the ARP mappings and explain how the attack is detected.

3. Learning Outcomes

- Explain how ARP maps IPv4 addresses to MAC addresses on a LAN.
- Inspect ARP caches on Linux and Windows systems.
- Identify ARP requests, legitimate replies and unsolicited or conflicting replies.
- Detect one IP address being associated with multiple MAC addresses.
- Capture and document a bounded ARP-poisoning simulation in a host-only network.
- Restore network state and recommend preventive controls.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- The live component is optional and must use instructor-assigned RFC1918 addresses on a host-only/internal virtual switch. Never poison a home, school, office, hotel or public Wi-Fi network.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Virtual machines	Kali attacker/analyst plus Linux or Windows victim	Create an isolated LAN scenario.
Network mode	Host-only/internal network	Prevent access to production networks.
Capture tools	Wireshark and tshark	Capture ARP packets.
Packet library	python3-scapy	Run authorized training scripts.
System tools	ip neigh, arp, ipconfig /all	Inspect ARP tables and interfaces.
Evidence	arp.pcap or student capture	Analyze known poisoning indicators.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab5/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab5
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

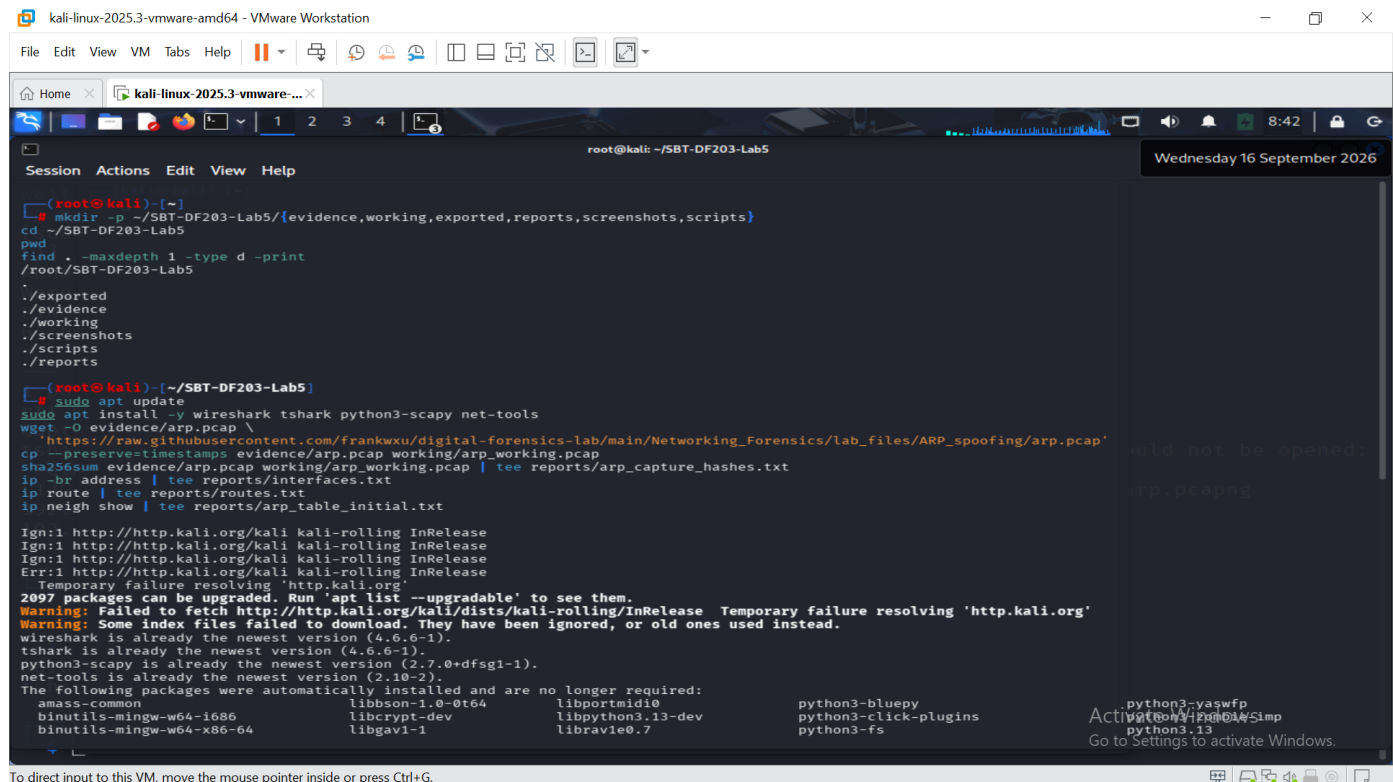
```

sudo apt update
sudo apt install -y wireshark tshark python3-scapy net-tools
wget -O evidence/arp.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-
lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap'
cp --preserve=timestamps evidence/arp.pcap working/arp_working.pcap
sha256sum evidence/arp.pcap working/arp_working.pcap | tee reports/arp_capture_hashes.txt
ip -br address | tee reports/interfaces.txt
ip route | tee reports/routes.txt
ip neigh show | tee reports/arp_table_initial.txt

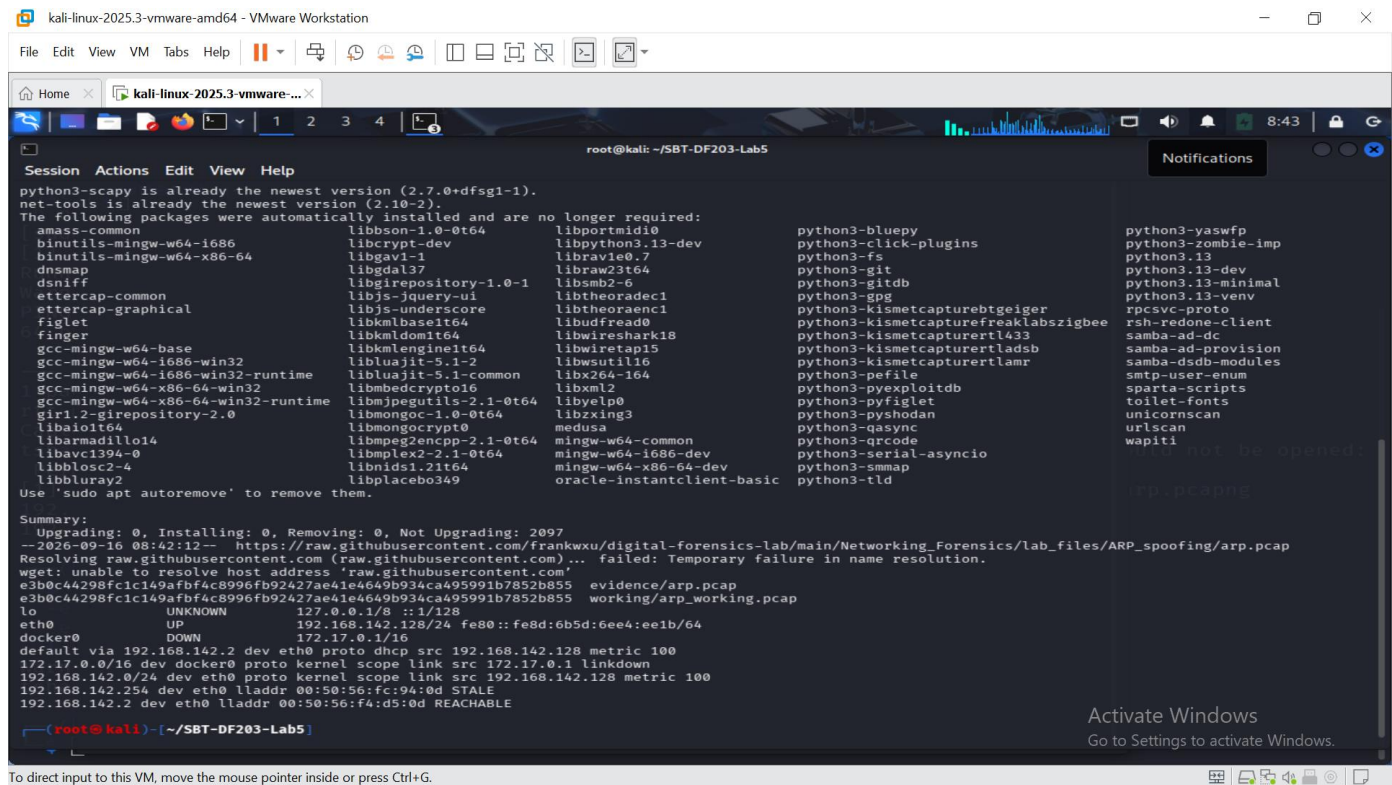
```

Screenshot required: Interface addresses, route/gateway, initial ARP table and evidence hashes.

Answer: See the attach screenshot



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



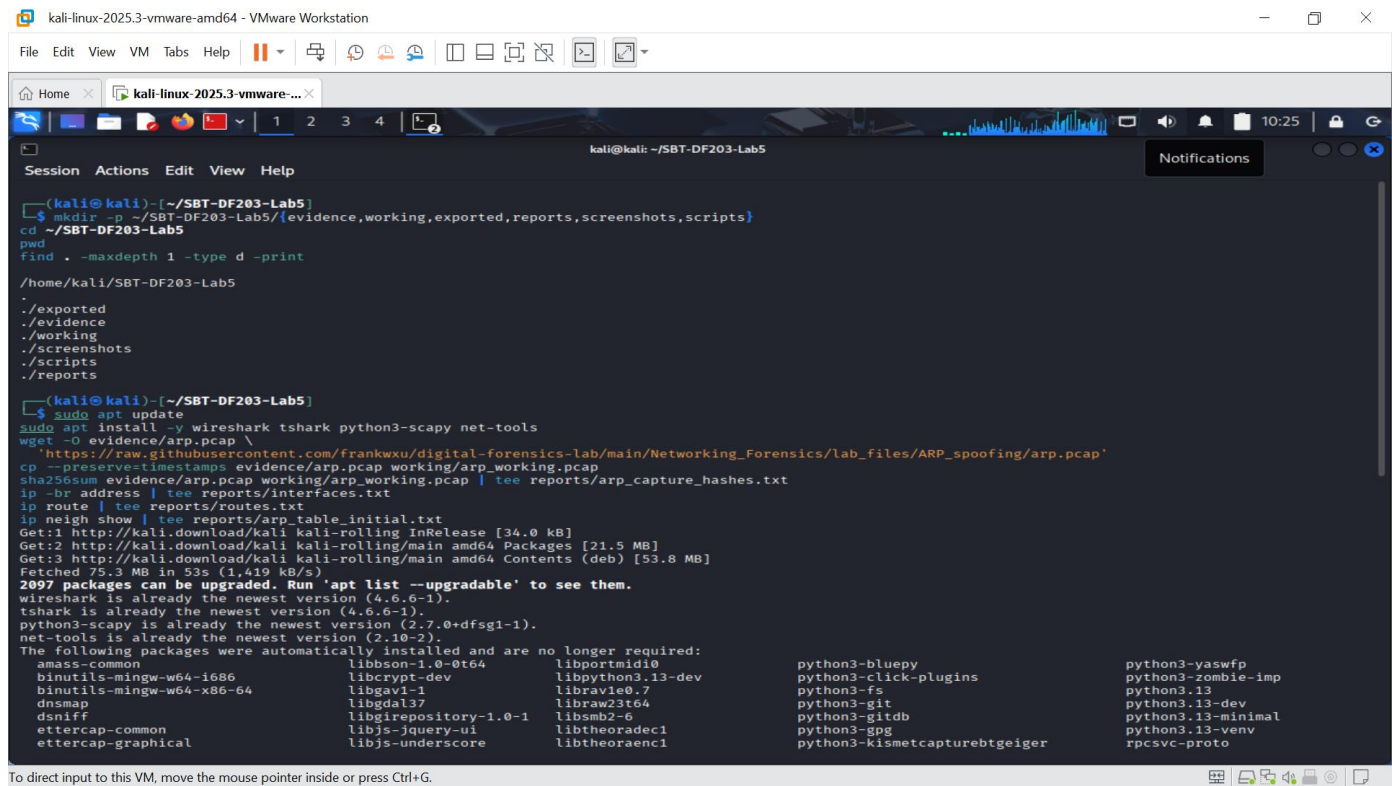
```
Session Actions Edit View Help
python3-scapy is already the newest version (2.7.0+dfsg1-1).
net-tools is already the newest version (2.10-2).
The following packages were automatically installed and are no longer required:
amass-common libbson-1.0-0t64 libportmidi0 python3-bluepy python3-click-plugins python3-yaswfp
binutils-mingw-w64-i686 libcrypt-dev libpython3.13-dev python3-click-plugins python3-zombie-imp
binutils-mingw-w64-x86-64 libgav1-1 libraw1e0.7 python3-fs python3.13 python3.13-dev
dnsmasq libgdal37 libraw23t64 python3-git python3.13-minimal python3.13-venv
ettercap-common libgirepository-1.0-1 libsm2-6 python3-gitdb python3.13-dev
ettercap-graphical libjs-jquery-ui libtheoradec1 python3-gpg python3-kismetcapturebtgeiger python3.13-venv
figlet libkmlbase1t64 libudfread0 python3-kismetcapturefreaklabszigbee rsh-redone-client
finger libkmldev1t64 libwireshare18 python3-kismetcapturertl433 samba-ad-dc
gcc-mingw-w64-base libkmlengine1t64 libwiretap15 python3-kismetcapturertldsb samba-ad-provision
gcc-mingw-w64-i686-win32 liblua5.1-2 libwsutil16 python3-kismetcapturertlamr samba-dsdb-modules
gcc-mingw-w64-i686-win32-runtime liblua5.1-common libx264-164 python3-pefile smtp-user-enum
gcc-mingw-w64-x86-64-win32-runtime libmbdcrpyto16 libxml2 python3-pyexploitdb sparta-scripts
gcc-mingw-w64-x86-64-win32-runtime libmjpegutils-2.1-0t64 libyelp0 python3-pyfiglet toilet-fonts
gir1.2-girepository-2.0 libmongoc-1.0-0t64 libzxing3 python3-pyshodan unicornscan
libaio1t64 medusa mingw-w64-common python3-qasync uriscan
libarmadillo14 libmp2encpp-2.1-0t64 mingw-w64-i686-dev python3-qrcode wapi1
libavc1304-0 libnids1.21t64 mingw-w64-x86-64-dev python3-serial-asyncio
libblosc2-4 libplacebo349 oracle-instantclient-basic python3-smmap
libbluray2

Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2097
--2026-09-16 08:42:12-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... failed: Temporary failure in name resolution.
wget: unable to resolve host address 'raw.githubusercontent.com'
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 evidence/arp.pcap
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 working/arp_working.pcap
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.142.128/24 fe80::fe8d:6b5d:6ee4:ee1b/64
docker0 DOWN 172.17.0.1/16
default via 192.168.142.2 dev eth0 proto dhcp src 192.168.142.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.142.0/24 dev eth0 proto kernel scope link src 192.168.142.128 metric 100
192.168.142.254 dev eth0 lladdr 00:50:56:fc:94:0d STALE
192.168.142.2 dev eth0 lladdr 00:50:56:f4:d5:0d REACHABLE

(kali@kali) [~/SBT-DF203-Lab5]
root@kali: ~/SBT-DF203-Lab5
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
Session Actions Edit View Help
(kali@kali) [~/SBT-DF203-Lab5]
$ mkdir -p ~/SBT-DF203-Lab5/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab5
pwd
find . -maxdepth 1 -type d -print

/home/kali/SBT-DF203-Lab5
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali) [~/SBT-DF203-Lab5]
$ sudo apt update
sudo apt install -y wireshark tshark python3-scapy net-tools
wget -O evidence/arp.pcap \
  "https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap"
cp --preserve-timestamps evidence/arp.pcap working/arp_working.pcap
sha256sum evidence/arp.pcap working/arp_working.pcap | tee reports/arp_capture_hashes.txt
ip -br address | tee reports/interfaces.txt
ip route | tee reports/routes.txt
ip neigh show | tee reports/arp_table_initial.txt
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.8 MB]
Fetched 75.3 MB in 53s (1,419 kB/s)
2097 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3-scapy is already the newest version (2.7.0+dfsg1-1).
net-tools is already the newest version (2.10-2).
The following packages were automatically installed and are no longer required:
amass-common libbson-1.0-0t64 libportmidi0 python3-bluepy python3-click-plugins python3-yaswfp
binutils-mingw-w64-i686 libcrypt-dev libpython3.13-dev python3-click-plugins python3-zombie-imp
binutils-mingw-w64-x86-64 libgav1-1 libraw1e0.7 python3-fs python3.13 python3.13-dev
dnsmasq libgdal37 libraw23t64 python3-git python3.13-minimal python3.13-venv
ettercap-common libgirepository-1.0-1 libsm2-6 python3-gitdb python3-kismetcapturebtgeiger python3.13-dev
ettercap-graphical libjs-jquery-ui libtheoradec1 python3-gpg python3-kismetcapturefreaklabszigbee rsh-redone-client
libkmlbase1t64 libudfread0 python3-kismetcapturertl433 samba-ad-dc
libkmldev1t64 libwireshare18 python3-kismetcapturertldsb samba-ad-provision
libkmlengine1t64 liblua5.1-2 libwsutil16 python3-kismetcapturertlamr samba-dsdb-modules
liblua5.1-common libx264-164 python3-pefile smtp-user-enum
libmbdcrpyto16 libxml2 python3-pyexploitdb sparta-scripts
libmjpegutils-2.1-0t64 libyelp0 python3-pyfiglet toilet-fonts
libmongoc-1.0-0t64 libzxing3 python3-pyshodan unicornscan
medusa mingw-w64-common python3-qasync uriscan
libmp2encpp-2.1-0t64 mingw-w64-i686-dev python3-qrcode wapi1
libnids1.21t64 mingw-w64-x86-64-dev python3-serial-asyncio
libplacebo349 oracle-instantclient-basic python3-smmap

(kali@kali) [~/SBT-DF203-Lab5]
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

kali@kali: ~/SBT-DF203-Lab5
CPU usage: 71.1%

Session Actions Edit View Help

ettercap-common          libjs-jquery-ui          libtheoraec1             python3-gpg              python3-kismetcapturebtgeiger  python3.13-venv
ettercap-graphical       libjs-underscore         libtheoraenc1            python3-kismetcapturefreaklabszigbee  rsh-redone-client
figlet                   libkmlbase1t64           libudfread0              python3-kismetcapturertl433            samba-ad-dc
finger                   libkmlengine1t64         libwiretap15             python3-kismetcaptureertladsb          samba-ad-provision
gcc-mingw-w64-base       gcc-mingw-w64-i686-win32 libluajit-5.1-2          python3-kismetcaptureertlamr          samba-dsdb-modules
gcc-mingw-w64-i686-win32-runtime  liblua5.1-common        libx264-164              python3-pefile              smtp-user-enum
gcc-mingw-w64-x86_64-win32  libmbedcrypto16         libxml2                  python3-pyexploitdb          sparta-scripts
gcc-mingw-w64-x86_64-win32-runtime  libmjpegutils-2.1-0t64  libyelp0                 python3-pyfiglet             toilet-fonts
git-2.43.0                libmongoc-1.0-0t64      medusa                   python3-pyshodan             unicornscan
libaio1t64               libmongocrypt0           mingw-w64-common         python3-qasync               python3-qrcode
libarmadillo14            libmpeg2encpp-2.1-0t64  mingw-w64-i686-dev       python3-serial-asyncio      python3-smmap
libavc1394-0              libmpeg2-2.1-0t64       mingw-w64-x86_64-dev     python3-tld
libblosc2-4               libnids1.21t64           oracle-instantclient-basic
libbluray2               libplacebo349

Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2097
--2026-09-15 10:19:41-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1072 (1.0K) [application/octet-stream]
Saving to: 'evidence/arp.pcap'

evidence/arp.pcap      100%[=====] 1.05K  --.-KB/s  in 0s

2026-09-15 10:19:42 (2.18 MB/s) - 'evidence/arp.pcap' saved [1072|1072]

342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e  evidence/arp.pcap
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e  working/arp_working.pcap
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.142.128/24 fe80::fe8d:6b5d:6ee4:ee1b/64
docker0 DOWN 172.17.0.1/16
default via 192.168.142.2 dev eth0 proto dhcp src 192.168.142.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.142.0/24 dev eth0 proto kernel scope link src 192.168.142.128 metric 100
192.168.142.254 dev eth0 lladdr 00:50:56:fc94:0d STALE
192.168.142.2 dev eth0 lladdr 00:50:56:f4:d5:0d REACHABLE

kali@kali: [~/SBT-DF203-Lab5]

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab5- Hussein Suleiman
Trainee name	HUSSEINI SULEIMAN
Date and time started	15 th September, 2026 – 03:20pm
Evidence file name(s)	arp.pcap
Source or generation method	Kali command line
Original SHA-256	342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e evidence/arp.pcap
Working-copy SHA-256	342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e working/arp_working.pcap
Analysis workstation/VM	kali@kali
Notes on any changes	nil

8. Part A - Observe Normal ARP Resolution

1. Identify the victim IP, gateway IP and your interface name.
2. Clear only the specific training neighbor entry where permitted.
3. Start an ARP-only capture, ping the gateway once and stop the capture.
4. Identify the broadcast request and unicast reply.


```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/SBT-DF203-Lab5
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab5]
$ IFACE=eth0
GATEWAY_IP=$(ip route | awk '/default/ {print $3; exit}')
sudo ip neigh flush "$GATEWAY_IP" dev "$IFACE"
sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w evidence/normal_arp.pcapng &
sleep 2
ping -c 1 "$GATEWAY_IP"
wait
ip neigh show | tee reports/arp_table_after_ping.txt
[sudo] password for kali:
[1] 270818
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
PING 192.168.142.2 (192.168.142.2) 56(84) bytes of data.
64 bytes from 192.168.142.2: icmp_seq=1 ttl=128 time=1.94 ms

--- 192.168.142.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.938/1.938/1.938/0.000 ms
Capturing on 'eth0'
tshark: The file to which the capture would be saved ("evidence/normal_arp.pcapng") could not be opened:
Permission denied.
[1] + exit 1 sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w evidence/normal_arp.pcapng
192.168.142.254 dev eth0 lladdr 00:50:56:fc:94:0d STALE
192.168.142.2 dev eth0 lladdr 00:50:56:f4:d5:0d REACHABLE

(kali@kali)-[~/SBT-DF203-Lab5]
$

```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

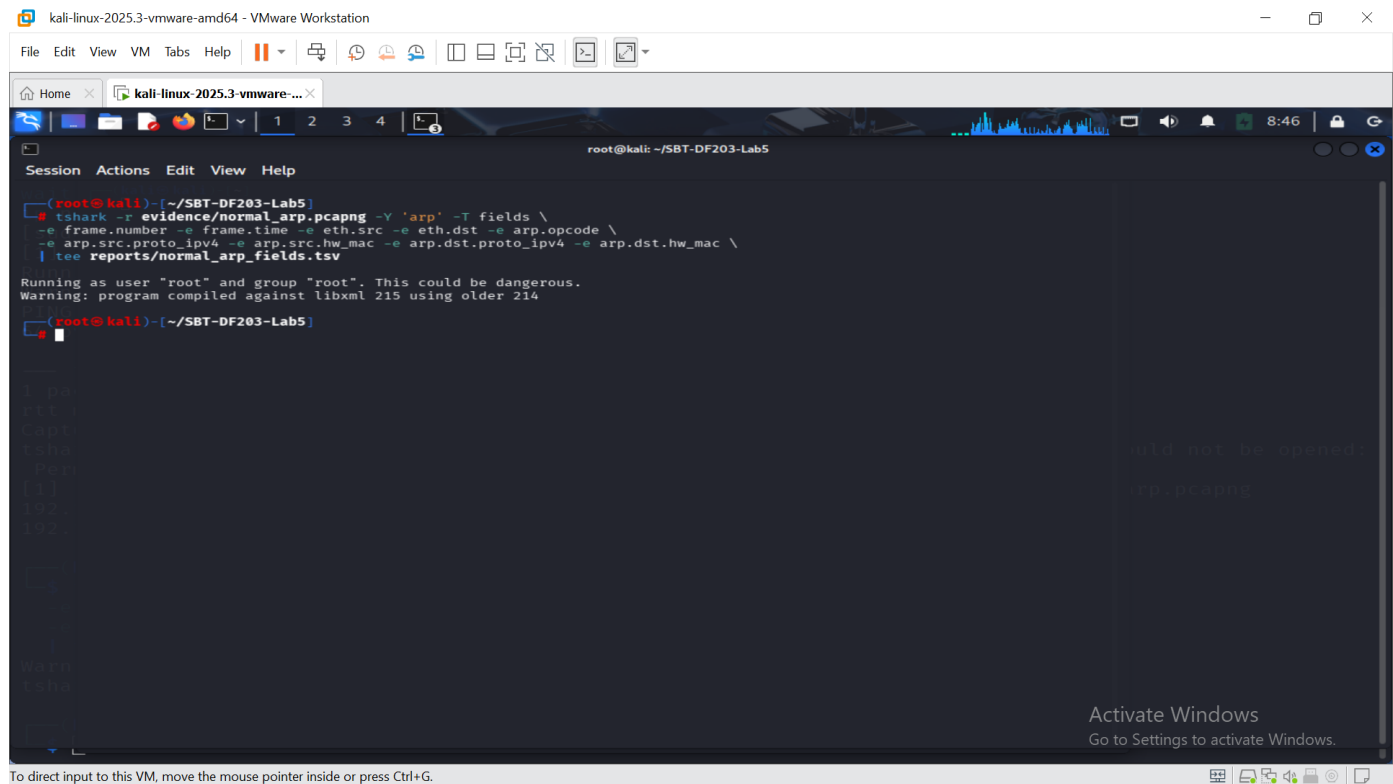
9. Part B - Analyze ARP Request and Reply Fields

```

tshark -r evidence/normal_arp.pcapng -Y 'arp' -T fields \
  -e frame.number -e frame.time -e eth.src -e eth.dst -e arp.opcode \
  -e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
  | tee reports/normal_arp_fields.tsv

```

Field	Normal Request	Normal Reply
Ethernet destination	ff:ff:ff:ff:ff:ff	00:0c:29:55:b0:67
ARP opcode	1	2
Sender protocol address 172.17.0.1	Requester IP: 192.168.142.128	Target IP 192.168.142.2
Sender hardware address 02:42:9e:bd:cd:b4	Requester MAC 00:0c:29:55:b0:67	Target MAC 00:00:00:00:00:00
Target protocol address	Target IP 192.168.142.2	Requester IP 192.168.142.128
Forensic interpretation	Who has target IP? The reply field	Target IP is at target MAC. 00:50:56:f4:d5:0d



10. Part C - Analyze the Supplied Poisoning Capture

```
PCAP=working/arp_working.pcap
# Inventory all ARP claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_replies.tsv

# Summarize IP-to-MAC claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
| sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt

# Look for gratuitous or unsolicited-looking replies
tshark -r "$PCAP" -Y 'arp.opcode==2 && eth.dst!=ff:ff:ff:ff:ff:ff' -T fields \
-e frame.number -e frame.time -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
| tee reports/unicast_arp_replies.tsv
```

```

kali@kali: ~/SBT-DF203-Lab5
$ PCAP=working/arp_working.pcap
# Inventory all ARP claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
  -e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
  | tee reports/arp_replies.tsv
Warning: program compiled against libxml 215 using older 214
4      1678591364.904123642      00:50:56:86:02:65      136.160.215.194 00:50:56
6      1678591369.943043799      00:50:56:86:02:65      136.160.215.15  00:50:56

kali@kali: ~/SBT-DF203-Lab5
$ # Summarize IP-to-MAC claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
  | sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt
Warning: program compiled against libxml 215 using older 214
1 136.160.215.194 00:50:56:86:02:65
1 136.160.215.15  00:50:56:86:02:65

kali@kali: ~/SBT-DF203-Lab5
$ # Look for gratuitous or unsolicited-looking replies
tshark -r "$PCAP" -Y 'arp.opcode==2 && eth.dst==ff:ff:ff:ff:ff:ff' -T fields \
  -e frame.number -e frame.time -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
  | tee reports/unicast_arp_replies.tsv
Warning: program compiled against libxml 215 using older 214
4      2023-03-11T22:22:44.904123642-0500      136.160.215.194 00:50:56:86:02:65      00:50:56:86:cb:f
6      2023-03-11T22:22:49.943043799-0500      136.160.215.15  00:50:56:86:02:65      00:50:56:86:02:6

```

```

kali@kali: ~/SBT-DF203-Lab5
$ PCAP=working/arp_working.pcap
# Inventory all ARP claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
  -e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
  | tee reports/arp_replies.tsv

# Summarize IP-to-MAC claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
  | sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt

# Look for gratuitous or unsolicited-looking replies
tshark -r "$PCAP" -Y 'arp.opcode==2 && eth.dst==ff:ff:ff:ff:ff:ff' -T fields \
  -e frame.number -e frame.time -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
  | tee reports/unicast_arp_replies.tsv
Warning: program compiled against libxml 215 using older 214
4      1678591364.904123642      00:50:56:86:02:65      136.160.215.194 00:50:56:86:02:65      136.160.215.15  00:50:56:86:cb:fc
6      1678591369.943043799      00:50:56:86:02:65      136.160.215.15  00:50:56:86:02:65      136.160.215.194 00:50:56:86:02:65
Warning: program compiled against libxml 215 using older 214
1 136.160.215.194 00:50:56:86:02:65
1 136.160.215.15  00:50:56:86:02:65
Warning: program compiled against libxml 215 using older 214
4      2023-03-11T22:22:44.904123642-0500      136.160.215.194 00:50:56:86:02:65      00:50:56:86:cb:fc
6      2023-03-11T22:22:49.943043799-0500      136.160.215.15  00:50:56:86:02:65      00:50:56:86:02:65

kali@kali: ~/SBT-DF203-Lab5
$

```

11. Part D - Optional Controlled Host-Only Simulation

Instructor Approval Required

Skip this section unless the virtual network is confirmed as host-only/internal and the victim, gateway and analyst IP addresses are assigned specifically for the lab.

- Record clean ARP tables on victim and gateway.
- Start ARP capture on the analyst VM.
- Run the instructor-provided arp.py for no more than 30 seconds.
- Capture the changed victim mapping and stop the script.
- Restore correct ARP mappings or restart the isolated VMs.

Example placeholders only - replace with instructor-assigned lab IPs

VICTIM_IP=192.168.56.20

GATEWAY_IP=192.168.56.1

IFACE=eth0

```
sudo tshark -i "$IFACE" -f 'arp' -a duration:40 -w evidence/controlled_arp_poison.pcapng &
sleep 2
```

In another terminal, run the instructor-provided script for <=30 seconds:

```
# sudo timeout 30 python3 scripts/arp.py "$VICTIM_IP" "$GATEWAY_IP"
```

wait

```
ip neigh show | tee reports/analyst_arp_table_after.txt
```

The screenshot shows a Kali Linux terminal window with the following content:

```

kali@kali: ~/SBT-DF203-Lab5
Session Actions Edit View Help
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_replies.tsv

# Summarize IP-to-MAC claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
| sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt

# Look for gratuitous or unsolicited-looking replies
tshark -r "$PCAP" -Y 'arp.opcode==2 && eth.dst==ff:ff:ff:ff:ff:ff' -T fields \
-e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
| tee reports/unicast_arp_replies.tsv
Warning: program compiled against libxml 215 using older 214
4 1678591364.904123642 00:50:56:86:02:65 136.160.215.194 00:50:56:86:02:65 136.160.215.15 00:50:56:86:cb:fc
6 1678591369.943043799 00:50:56:86:cb:fc 136.160.215.15 00:50:56:86:02:65 136.160.215.194 00:50:56:86:cb:fc
Warning: program compiled against libxml 215 using older 214
1 136.160.215.194 00:50:56:86:02:65
1 136.160.215.15 00:50:56:86:cb:fc
Warning: program compiled against libxml 215 using older 214
4 2023-03-11T22:22:44.904123642-0500 136.160.215.194 00:50:56:86:02:65 00:50:56:86:cb:fc
6 2023-03-11T22:22:49.943043799-0500 136.160.215.15 00:50:56:86:cb:fc

(kali@kali)-[~/SBT-DF203-Lab5]
$ # Example placeholders only - replace with instructor-assigned lab IPs
VICTIM_IP=192.168.56.20
GATEWAY_IP=192.168.56.1
IFACE=eth0

(kali@kali)-[~/SBT-DF203-Lab5]
$ sudo tshark -i "$IFACE" -f 'arp' -a duration:40 -w evidence/controlled_arp_poison.pcapng &
sleep 2
# In another terminal, run the instructor-provided script for <=30 seconds:
# sudo timeout 30 python3 scripts/arp.py "$VICTIM_IP" "$GATEWAY_IP"

[1] 279493
[sudo] password for kali:
[1] + suspended (tty input) sudo tshark -i "$IFACE" -f 'arp' -a duration:40 -w

(kali@kali)-[~/SBT-DF203-Lab5]
$ ip neigh show | tee reports/analyst_arp_table_after.txt
192.168.142.254 dev eth0 lladdr 00:50:56:fc:94:0d STALE
192.168.142.2 dev eth0 lladdr 00:50:56:f4:d5:0d STALE

(kali@kali)-[~/SBT-DF203-Lab5]
$

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12. Part E - Detection Logic and Timeline

Time	Claimed IP	Claimed MAC	Target	Request Seen First?	Assessment
------	------------	-------------	--------	---------------------	------------

Time	Claimed IP	Claimed MAC	Target	Request Seen First?	Assessment
1678591364.904123642	136.160.215.194	00:50:56:86:02:65	136.160.215.194	00:50:56:86:02:65	The mac address changes
2023-03-11T22:22:49.943043799-0500	136.160.215.15	00:50:56:86:cb:fc	136.160.215.194	00:50:56:86:cb:fc	
	192.168.142.254 dev eth0	00:50:56:fc:94:0d	136.160.215.194	00:50:56:86:02:65	
	192.168.142.2	00:50:56:f4:d5:0d	192.168.142.254	00:50:56:fc:94:0d	

- Flag a gateway IP that changes from its legitimate MAC to the analyst/attacker MAC.
- Flag repeated ARP replies without a corresponding recent request.
- Flag one MAC claiming both victim and gateway IP addresses.
- Correlate ARP evidence with packet forwarding, DNS/HTTP anomalies and endpoint ARP-cache changes.

13. Restoration and Prevention

```
# Stop any training script first, then clear learned entries on the lab VM
sudo ip neigh flush all
# Renew/reconnect the isolated interface as appropriate
ip neigh show

# Confirm no ARP-poisoning process remains
ps aux | grep -E '[a]rp.py|[s]capy' | tee reports/process_check.txt
```

- Use Dynamic ARP Inspection with DHCP snooping on managed switches where available.
- Segment networks, enable port security and monitor gateway IP-to-MAC consistency.
- Use encrypted application protocols so interception does not expose plaintext.
- Deploy ARP monitoring tools and correlate alerts with switch and endpoint logs.

Required Screenshots Checklist

- ☐ Initial interface, route and ARP table.
- ☐ Normal broadcast ARP request.
- ☐ Normal ARP reply.
- ☐ Supplied capture hash.
- ☐ ARP reply inventory.
- ☐ IP-to-MAC claim summary.
- ☐ Conflicting gateway MAC evidence.
- ☐ Clean versus poisoned ARP table comparison.
- ☐ Controlled capture, if authorized.
- ☐ Restored ARP table and process check.

Student Report Template

10. Executive summary and authorization.
11. Network diagram and assigned IP/MAC addresses.
12. Evidence integrity.
13. Normal ARP behaviour.
14. Poisoning indicators and timeline.
15. ARP-table comparison.
16. Impact assessment.
17. Restoration and preventive controls.

18. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab5__Husseini Suleiman.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Safety and environment	15	Host-only scope and restoration documented.
Normal ARP analysis	15	Request/reply fields and ARP-cache behaviour.
Poisoning detection	25	Conflicting claims, unsolicited replies and timeline.
Controlled evidence or supplied PCAP analysis	20	Complete, repeatable capture analysis.
Mitigation and restoration	15	Correct cleanup and preventive controls.
Report quality	10	Clear diagram, screenshots and conclusions.